Matplotlib

- Library used for graphical representation of values
- Installation:
  - pip install matplotlib --> import matplotlib as plt
- Suppose we have to populate our previous lists:
  - import matplotlib.pyplot as plt
  - plt.plot(Convert)
  - plt.show()

Various charts or type of charts available in Matplotlib library:

- Histogram: Measure frequency distribution of data
- Area/ Bar charts: Represent no. of observation for different categories
- Pie Charts: Represent the percentage data by each category

In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [4]:
```
Data = {'Year': [1920,1930,1940,1950,1960,1970,1980,1990,2000, 2010, 2020],
        'Exchange Rate': [65, 69, 71, 64, 62, 59, 72, 71, 75, 78, 81]}
```

In [5]:
```
Data
```

Out[5]:
```
{'Year': [1920, 1930, 1940, 1950, 1960, 1970, 1980, 1990, 2000, 2010, 2020],
 'Exchange Rate': [65, 69, 71, 64, 62, 59, 72, 71, 75, 78, 81]}
```

In [6]:
```
type(Data)
```

Out[6]:
```
dict
```

In [8]:
```
df = pd.DataFrame(Data)
```

In [9]:
```
df
```

Out[9]:

	Year	Exchange Rate
0	1920	65
1	1930	69
2	1940	71
3	1950	64
4	1960	62
5	1970	59
6	1980	72
7	1990	71
8	2000	75
9	2010	78
10	2020	81

In [17]:
```
df = pd.DataFrame(Data, columns = ['Year', 'Exchange Rate'])
# In some cases not able to identify columns so its good job to mention the colu
```

In [18]:
```
df
```

Out[18]:

	Year	Exchange Rate
0	1920	65
1	1930	69
2	1940	71
3	1950	64
4	1960	62
5	1970	59
6	1980	72
7	1990	71
8	2000	75
9	2010	78
10	2020	81

In [19]:
```
df = pd.DataFrame(Data, columns = ['Year', 'Ex Rate'])
# Dectect the name exactly as defined inside the dictionary else keep it empty l
```

In [20]:
```
df
```

Out[20]:

	Year	Ex Rate
0	1920	NaN
1	1930	NaN
2	1940	NaN
3	1950	NaN
4	1960	NaN
5	1970	NaN
6	1980	NaN
7	1990	NaN
8	2000	NaN
9	2010	NaN
10	2020	NaN

In [21]:
```
df = pd.DataFrame(Data, columns = ['Year', 'Exchange Rate'])
```

In [23]:
```
df
```

Out[23]:

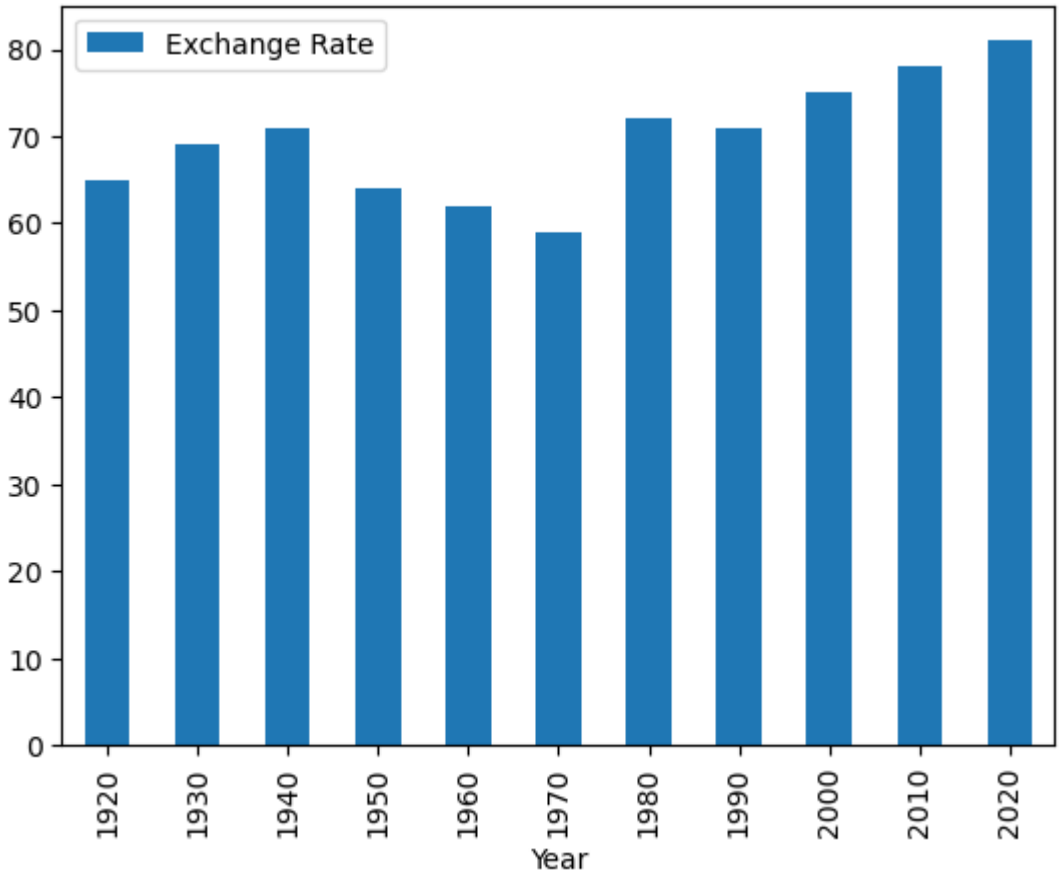
	Year	Exchange Rate
0	1920	65
1	1930	69
2	1940	71
3	1950	64
4	1960	62
5	1970	59
6	1980	72
7	1990	71
8	2000	75
9	2010	78
10	2020	81

In [24]:
```
type(df) # It is panda data frame
```

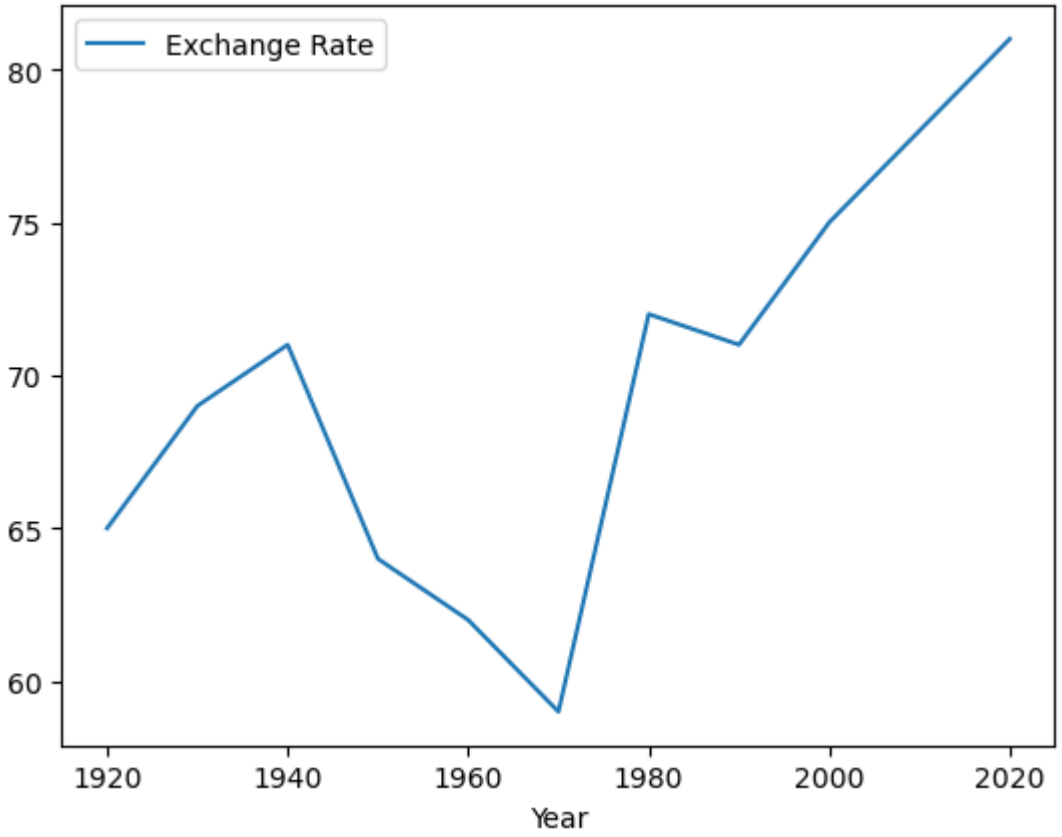
Out[24]:
```
pandas.core.frame.DataFrame
```

Instead of this if you want to use file then: df = pd.read_csv('filename.csv') df = pd.read_excel('filename.excel')

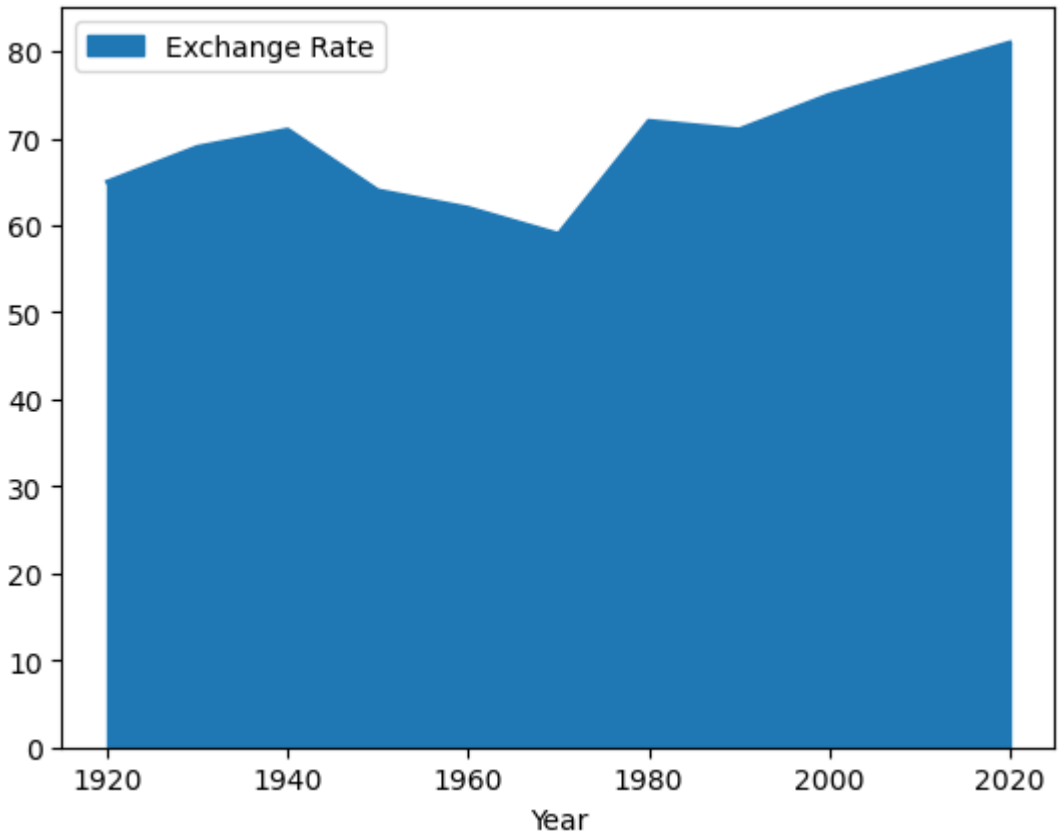
In [26]:
```
df.plot(x = 'Year', y = 'Exchange Rate', kind = 'bar')
plt.show()
```



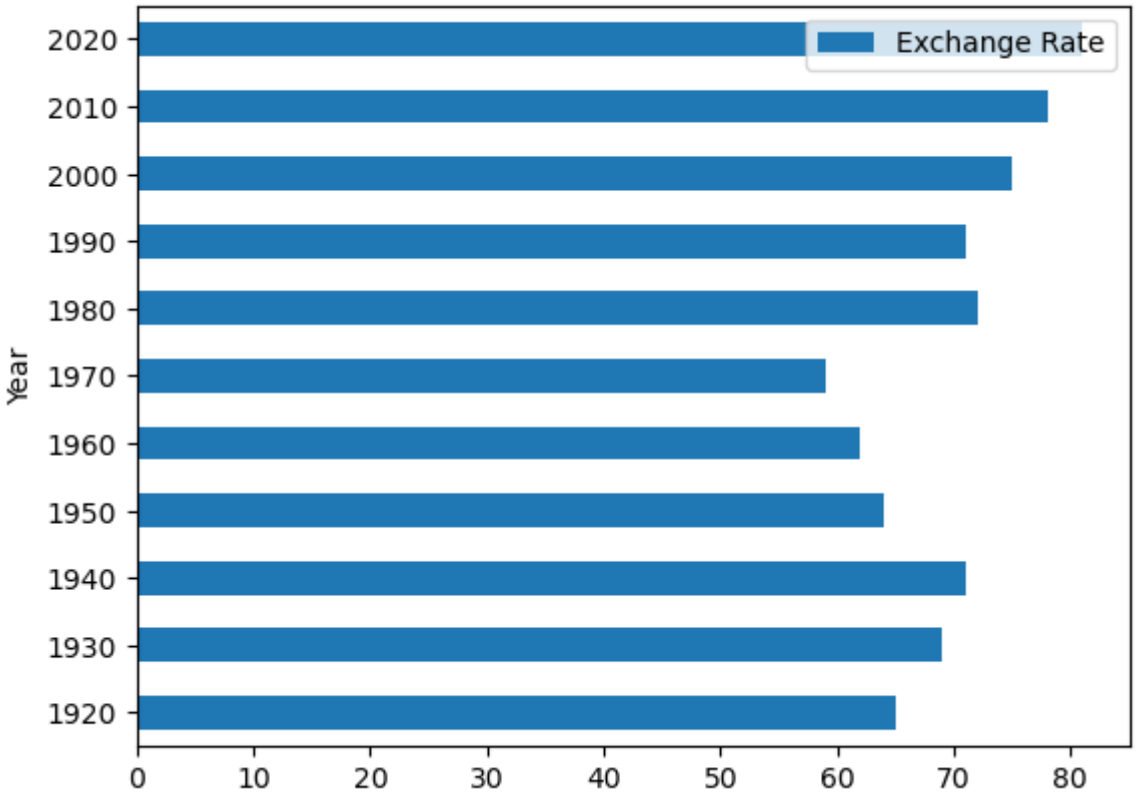
In [27]:
```
df.plot(x = 'Year', y = 'Exchange Rate', kind = 'line')
plt.show()
```



In [28]:
```
df.plot(x = 'Year', y = 'Exchange Rate', kind = 'area')
plt.show()
```

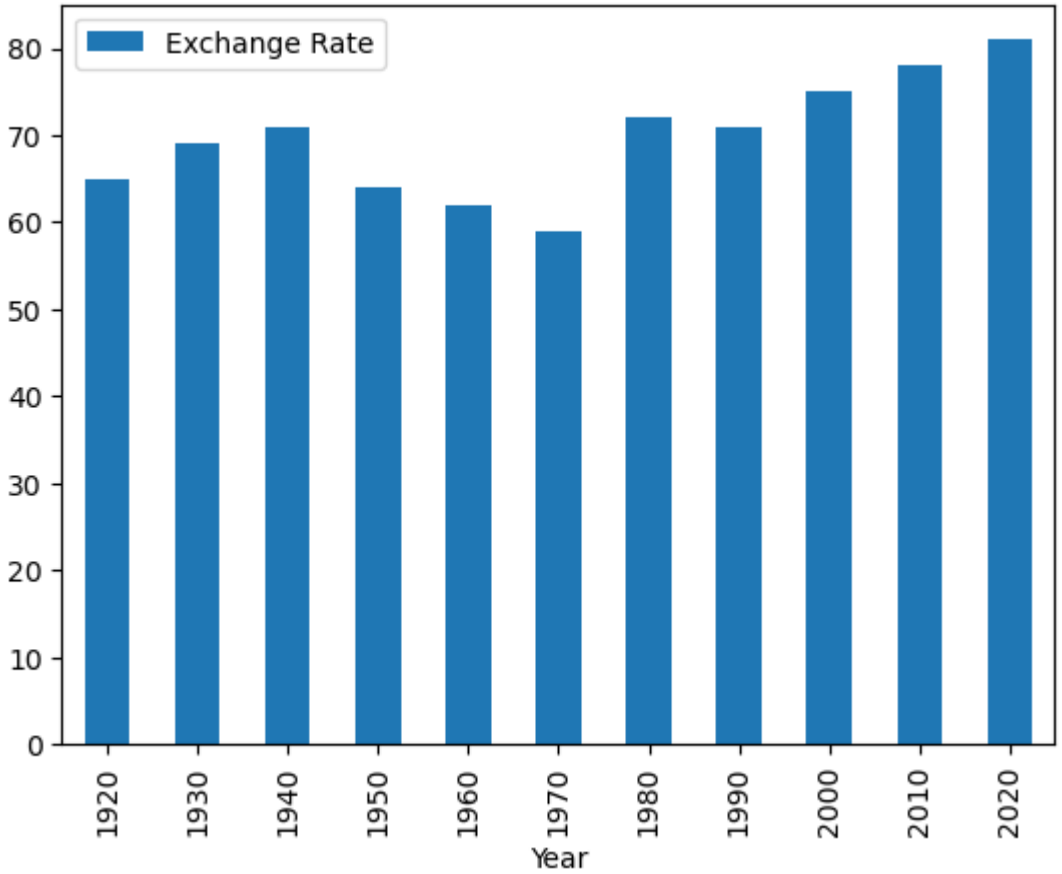


In [30]:
```
df.plot(x = 'Year', y = 'Exchange Rate', kind = 'barh')  # For horizontal bar ch
plt.show()
```

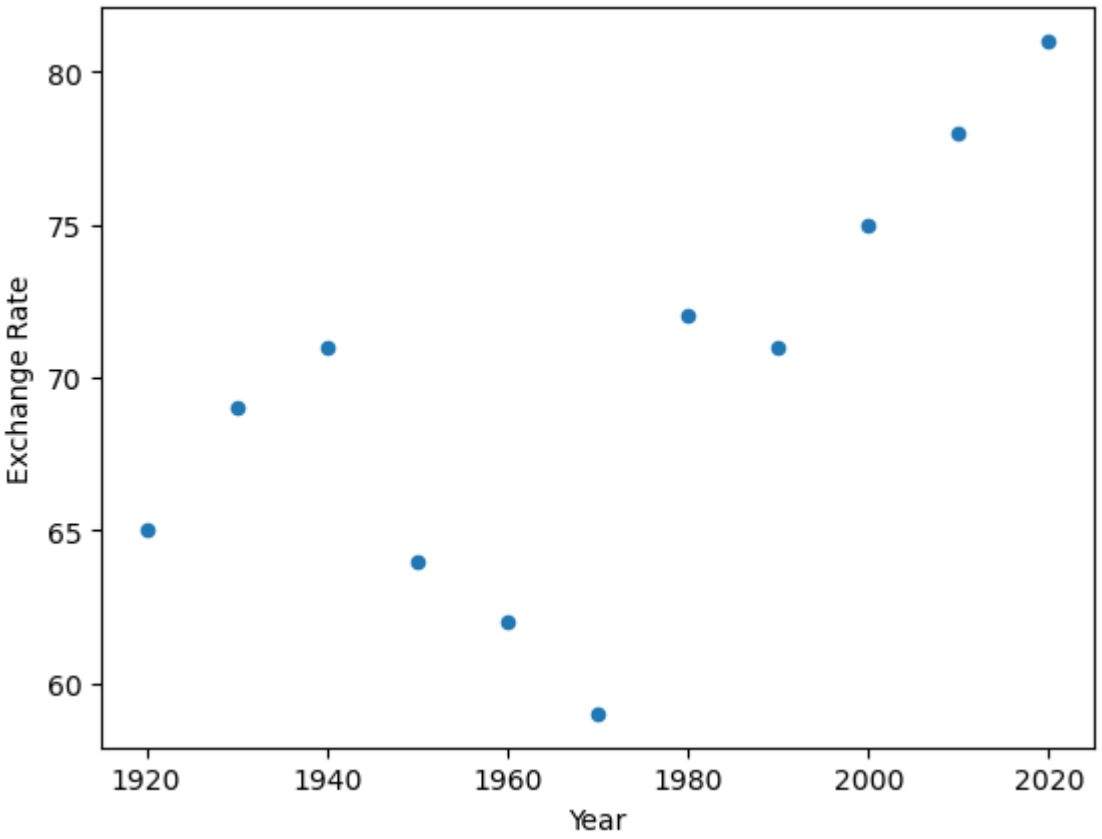


In [31]:
```
Data = {'Year': [1920,1930,1940,1950,1960,1970,1980,1990,2000, 2010, 2020],
        'Exchange Rate': [65, 69, 71, 64, 62, 59, 72, 71, 75, 78, 81]}

df = pd.DataFrame(Data, columns = ['Year', 'Exchange Rate'])
df.plot(x = 'Year', y = 'Exchange Rate', kind = 'bar')
```

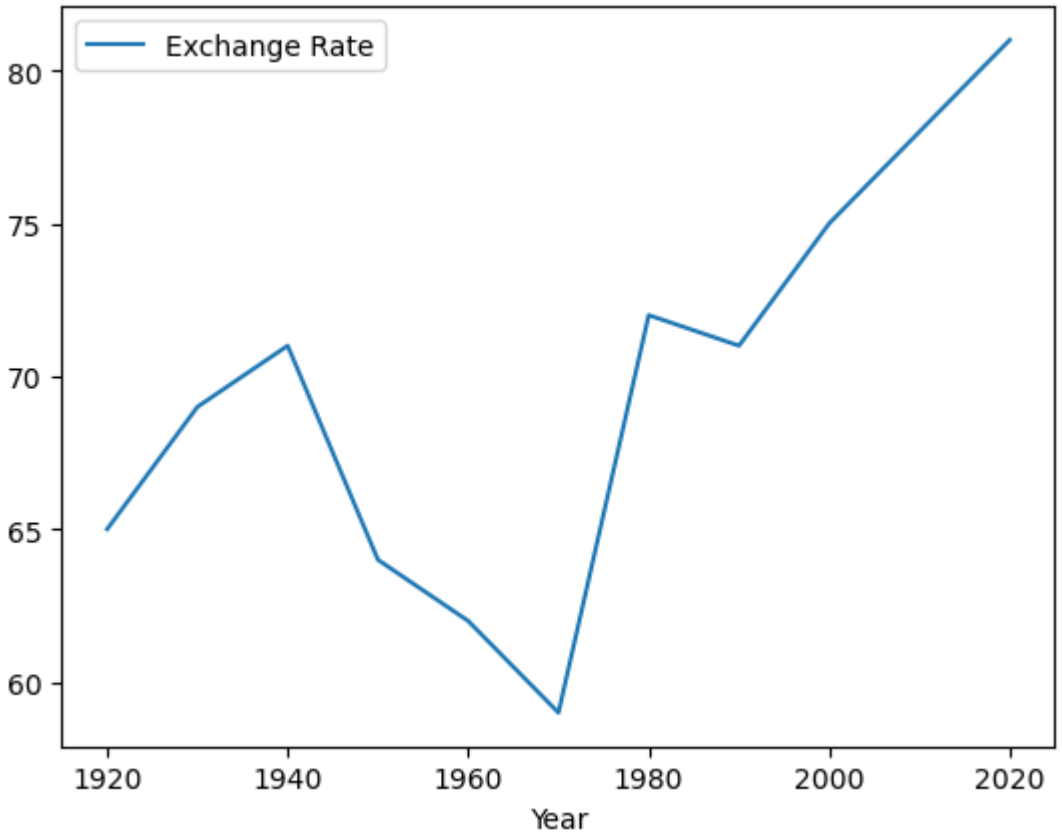
```
plt.show()
```



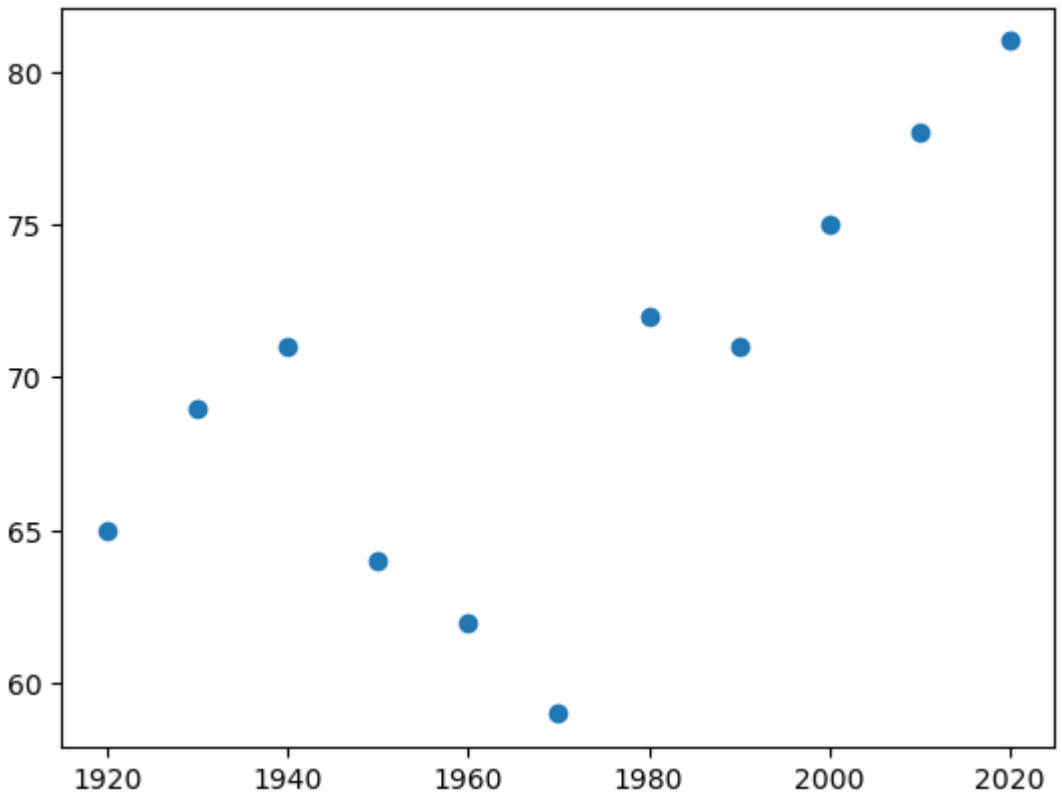
In [32]:
```
Data = {'Year': [1920,1930,1940,1950,1960,1970,1980,1990,2000, 2010, 2020],
        'Exchange Rate': [65, 69, 71, 64, 62, 59, 72, 71, 75, 78, 81]}

df = pd.DataFrame(Data, columns = ['Year', 'Exchange Rate'])
df.plot(x = 'Year', y = 'Exchange Rate', kind = 'scatter')
plt.show()
```



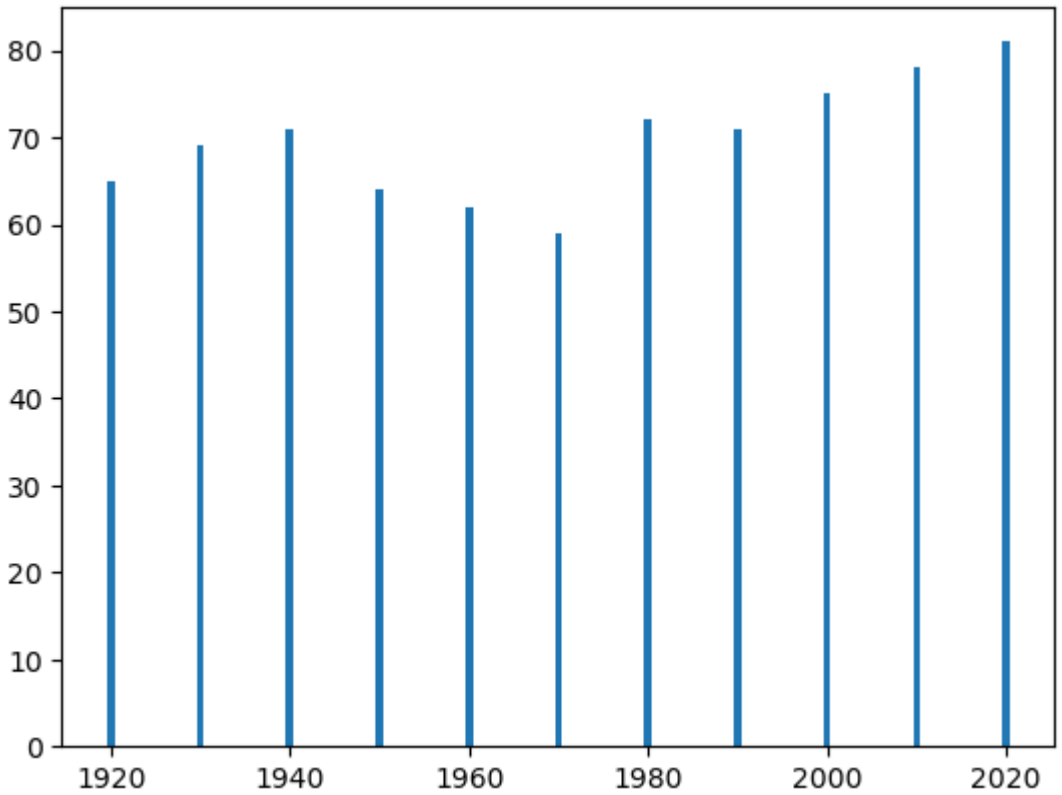
In [33]:
```
Data = {'Year': [1920,1930,1940,1950,1960,1970,1980,1990,2000, 2010, 2020],
        'Exchange Rate': [65, 69, 71, 64, 62, 59, 72, 71, 75, 78, 81]}

df = pd.DataFrame(Data, columns = ['Year', 'Exchange Rate'])
df.plot(x = 'Year', y = 'Exchange Rate')
plt.show()
```



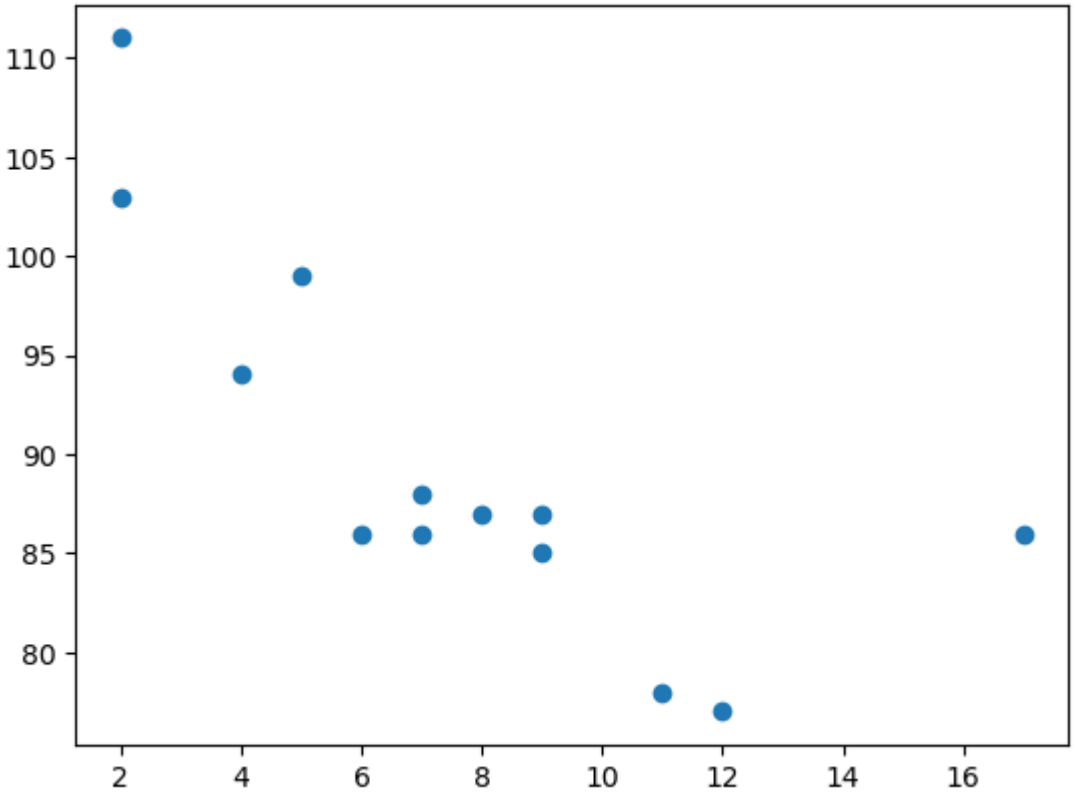
In [35]:
```
plt.scatter(df['Year'], df['Exchange Rate'])
plt.show()
```



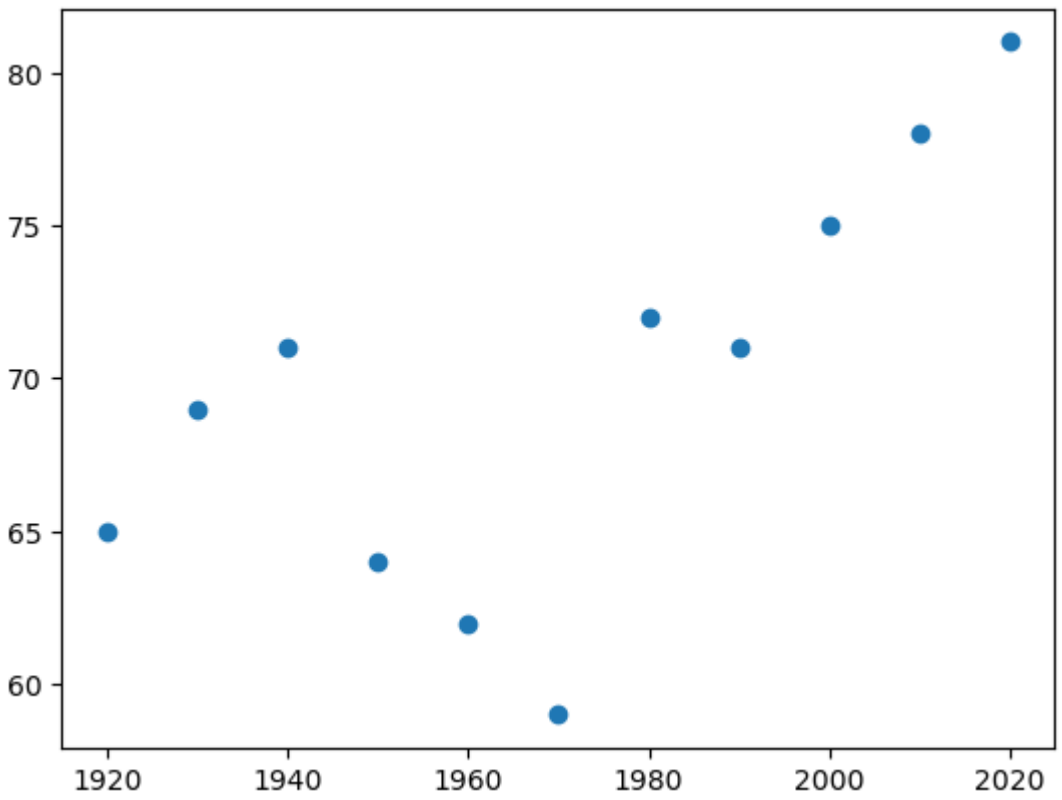
In [37]:
```
Data = {'Year': [1920,1930,1940,1950,1960,1970,1980,1990,2000, 2010, 2020],
        'Exchange Rate': [65, 69, 71, 64, 62, 59, 72, 71, 75, 78, 81]}

df = pd.DataFrame(Data, columns = ['Year', 'Exchange Rate'])
plt.bar(df['Year'], df['Exchange Rate'])
plt.show()
```



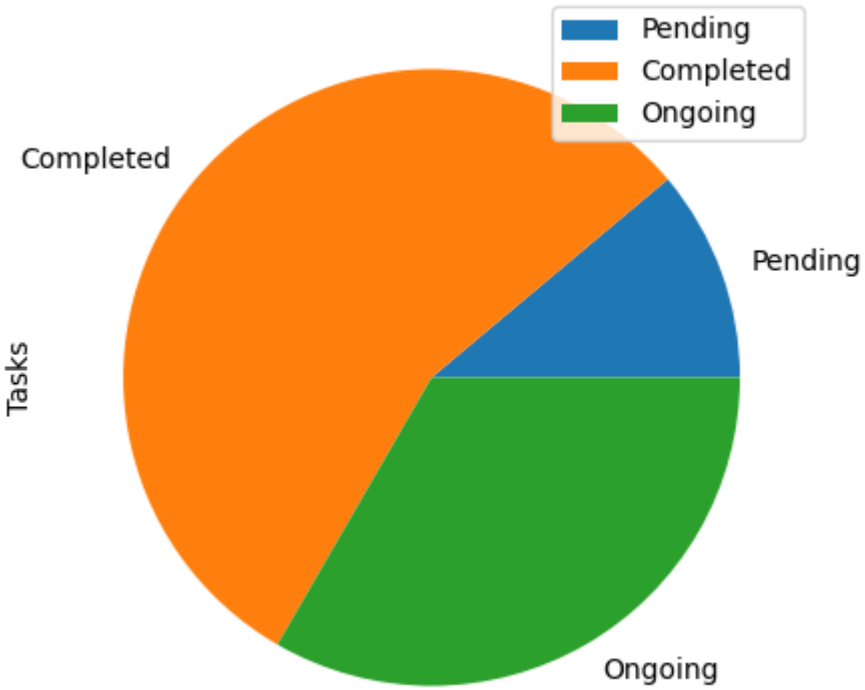
In [38]:
```
import matplotlib.pyplot as plt
import numpy as np

x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])

plt.scatter(x, y)
plt.show()
```



In [39]:
```
Data = {'Year': [1920,1930,1940,1950,1960,1970,1980,1990,2000, 2010, 2020],
        'Exchange Rate': [65, 69, 71, 64, 62, 59, 72, 71, 75, 78, 81]}

df = pd.DataFrame(Data, columns = ['Year', 'Exchange Rate'])
plt.scatter(df.Year, df['Exchange Rate'])
plt.show()
```

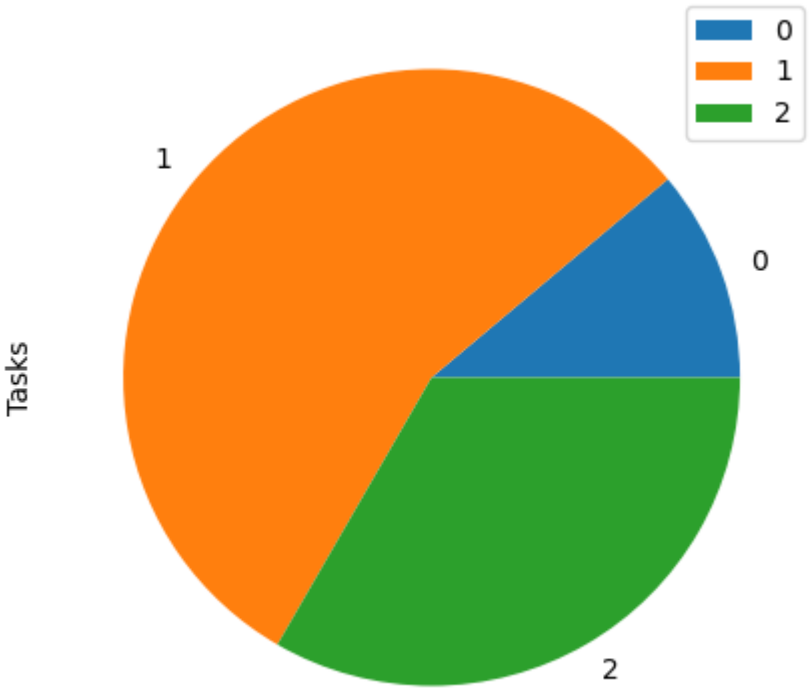


In [40]:
```
Data = {'Tasks': [100,500,300]}
df = pd.DataFrame(Data, columns = ['Tasks'], index = ['Pending', 'Completed', 'O

df.plot.pie(y='Tasks', figsize=(5,5))
plt.show()
```



In [44]:
```
Data = {'Tasks': [100,500,300]}
df = pd.DataFrame(Data, columns = ['Tasks'])

df.plot.pie(y='Tasks', figsize=(5,5))
```

```
plt.show()
```



In [45]:
```
df   # Default indexes 0,1 and 2
```

Out[45]:

	Tasks
0	100
1	500
2	300

In [46]:
```
churn_df = pd.read_csv('Churn_Modelling.csv')
```

In [49]:
```
churn_df
```

Out[49]:

	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenur
0	1	15634602	Hargrave	619	France	Female	42	
1	2	15647311	Hill	608	Spain	Female	41	
2	3	15619304	Onio	502	France	Female	42	
3	4	15701354	Boni	699	France	Female	39	
4	5	15737888	Mitchell	850	Spain	Female	43	
5	6	15678901	Smith	720	Germany	Male	35	
6	7	15623456	Johnson	645	France	Male	50	
7	8	15765432	Williams	580	Spain	Female	28	
8	9	15698765	Brown	610	Germany	Male	45	
9	10	15712345	Jones	675	France	Female	38	
10	11	15654321	Garcia	690	Spain	Male	55	
11	12	15687654	Miller	605	Germany	Female	33	
12	13	15743210	Davis	730	France	Female	47	
13	14	15621098	Rodriguez	540	Spain	Male	40	
14	15	15789012	Martinez	600	Germany	Female	36	
15	16	15634567	Hernandez	625	France	Male	29	
16	17	15678943	Lopez	710	Spain	Female	52	
17	18	15765489	Gonzalez	555	Germany	Male	48	
18	19	15612378	Wilson	640	France	Female	31	
19	20	15743298	Anderson	660	Spain	Male	44	

In [50]:
```
churn_df.head(5)
```

Out[50]:

	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure
0	1	15634602	Hargrave	619	France	Female	42	2
1	2	15647311	Hill	608	Spain	Female	41	1
2	3	15619304	Onio	502	France	Female	42	8
3	4	15701354	Boni	699	France	Female	39	1
4	5	15737888	Mitchell	850	Spain	Female	43	2

In [51]:
```
churn_df.tail(5)
```

Out[51]:

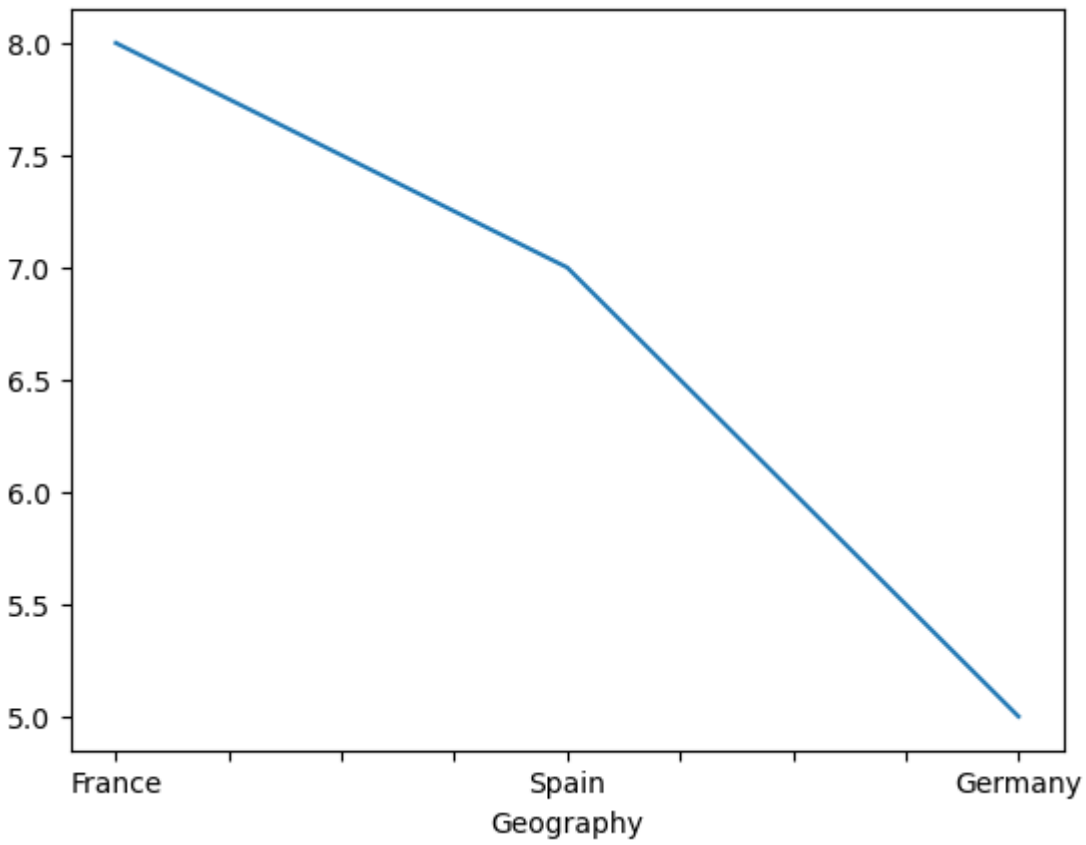
	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenur
15	16	15634567	Hernandez	625	France	Male	29	
16	17	15678943	Lopez	710	Spain	Female	52	
17	18	15765489	Gonzalez	555	Germany	Male	48	
18	19	15612378	Wilson	640	France	Female	31	
19	20	15743298	Anderson	660	Spain	Male	44	

In [54]:
```
churn_df.Geography.value_counts()
```

Out[54]:
```
Geography
France     8
Spain      7
Germany    5
Name: count, dtype: int64
```

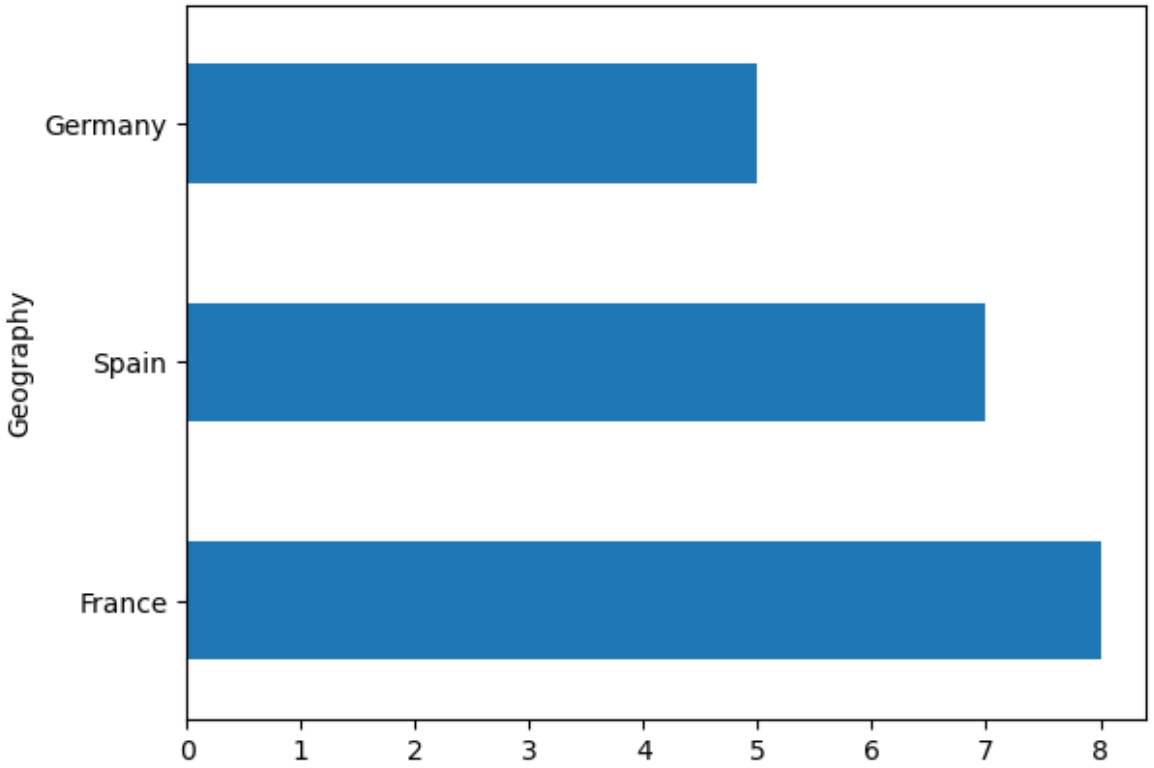
In [55]:
```
churn_df.Geography.value_counts().plot()
```

Out[55]:
```
<Axes: xlabel='Geography'>
```



In [53]:
```
churn_df.Geography.value_counts().plot(kind='barh')  # To find Total number reco
```

Out[53]:
```
<Axes: ylabel='Geography'>
```



In [59]:
```
churn_df['Exited'].value_counts()       # Exited- People left the company
```

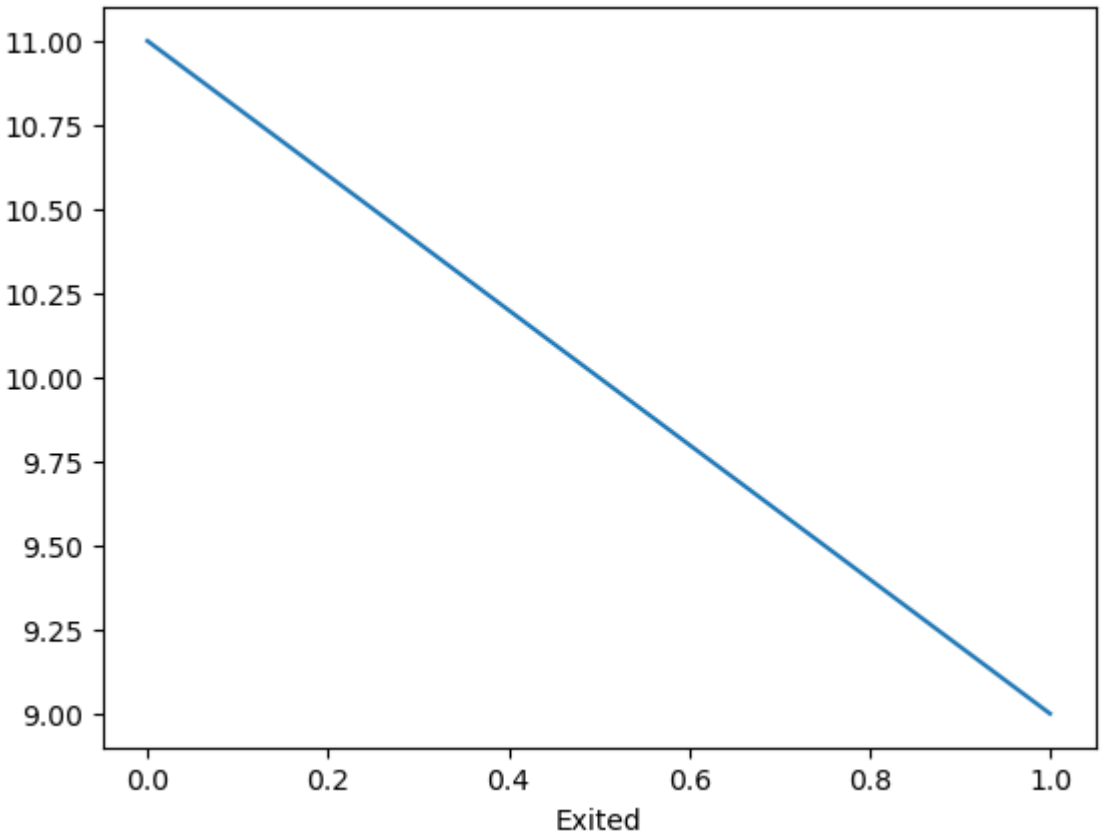
Out[59]:
```
Exited
0    11
1     9
Name: count, dtype: int64
```

In [61]:
```
churn_df['Exited'].value_counts()/len(churn_df)*100    # Percentage = len( ... )
```

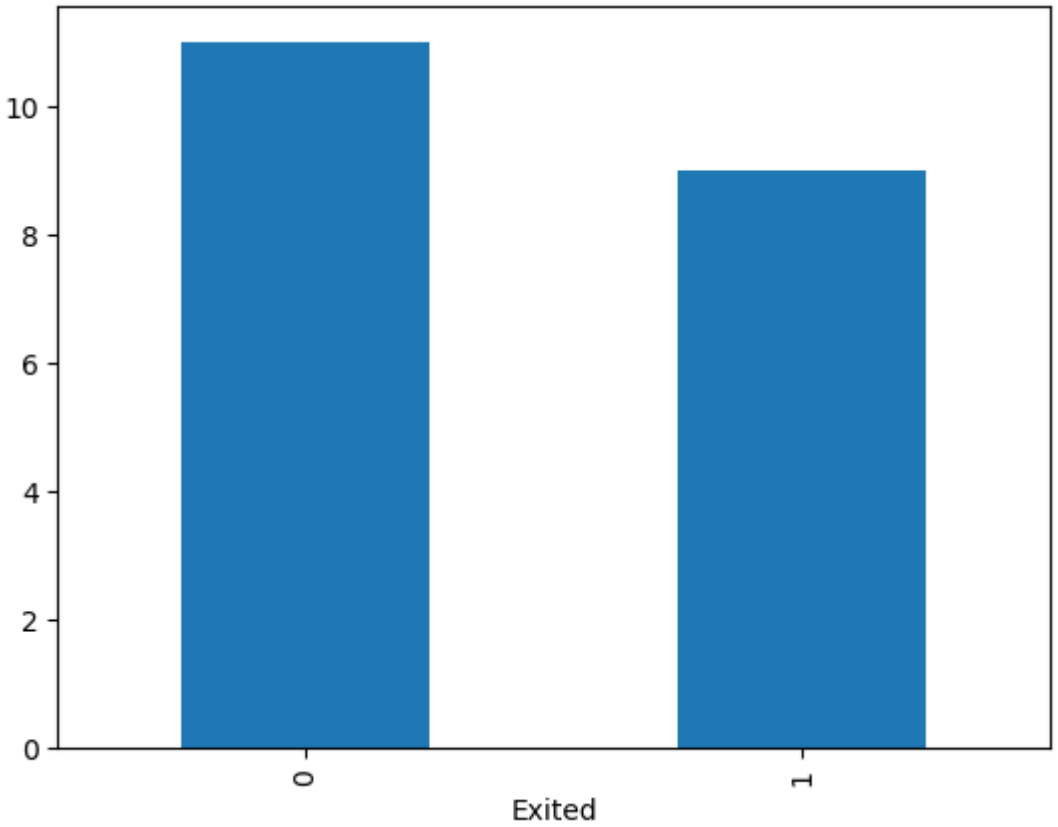
Out[61]:
```
Exited
0    55.0
1    45.0
Name: count, dtype: float64
```

In [58]:
```
churn_df['Exited'].value_counts().plot()
```

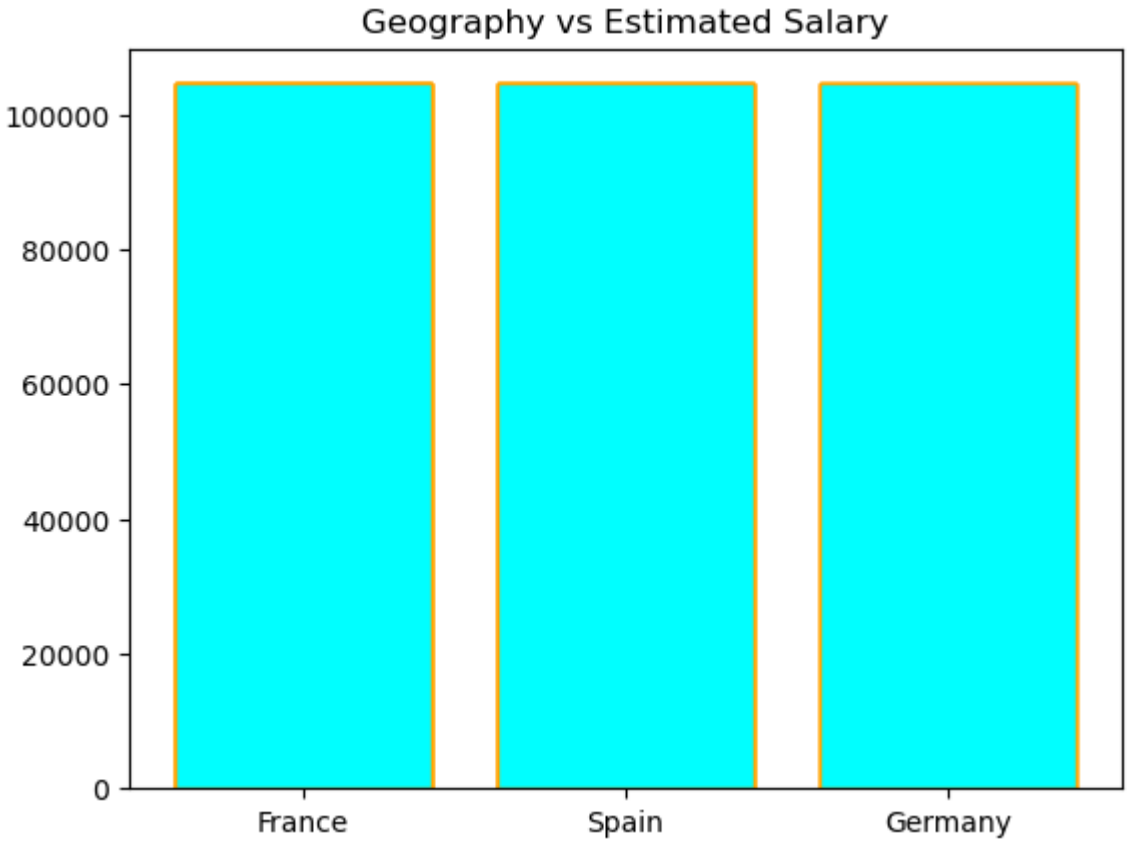
Out[58]:
```
<Axes: xlabel='Exited'>
```



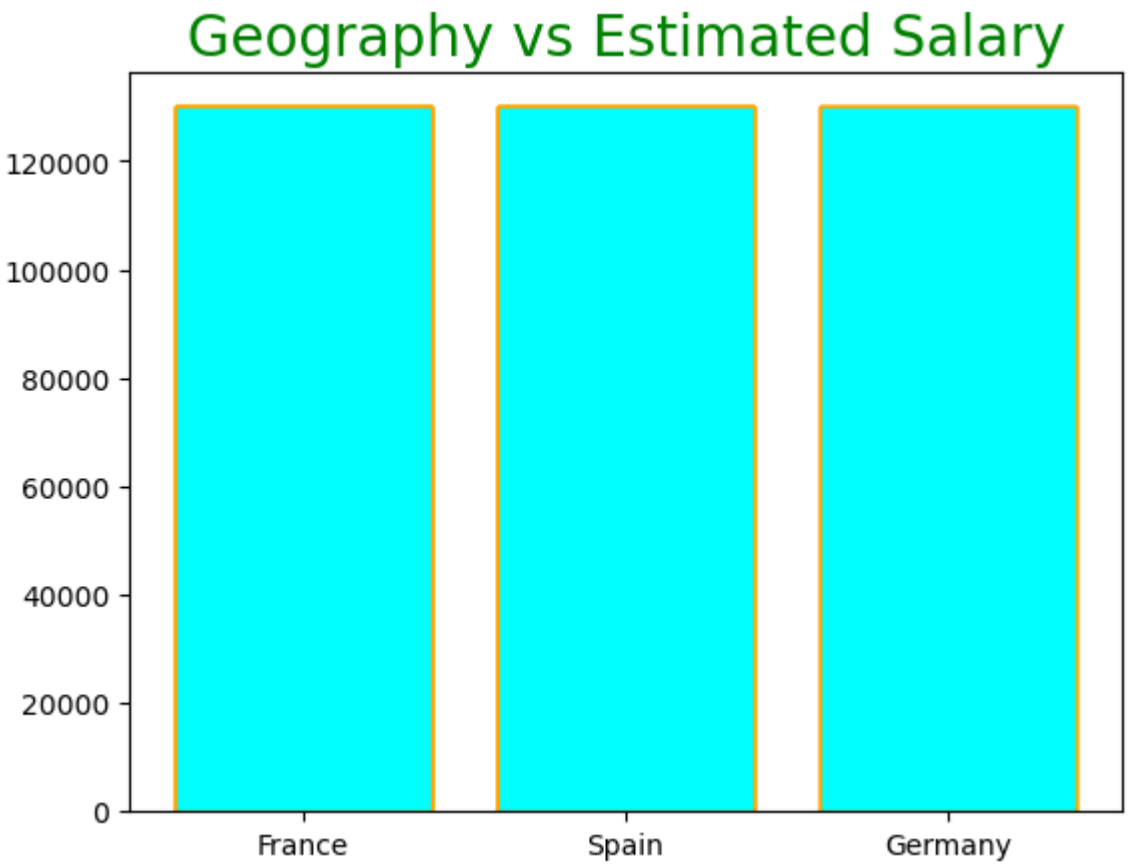
In [56]:
```
churn_df['Exited'].value_counts().plot(kind='bar');
```



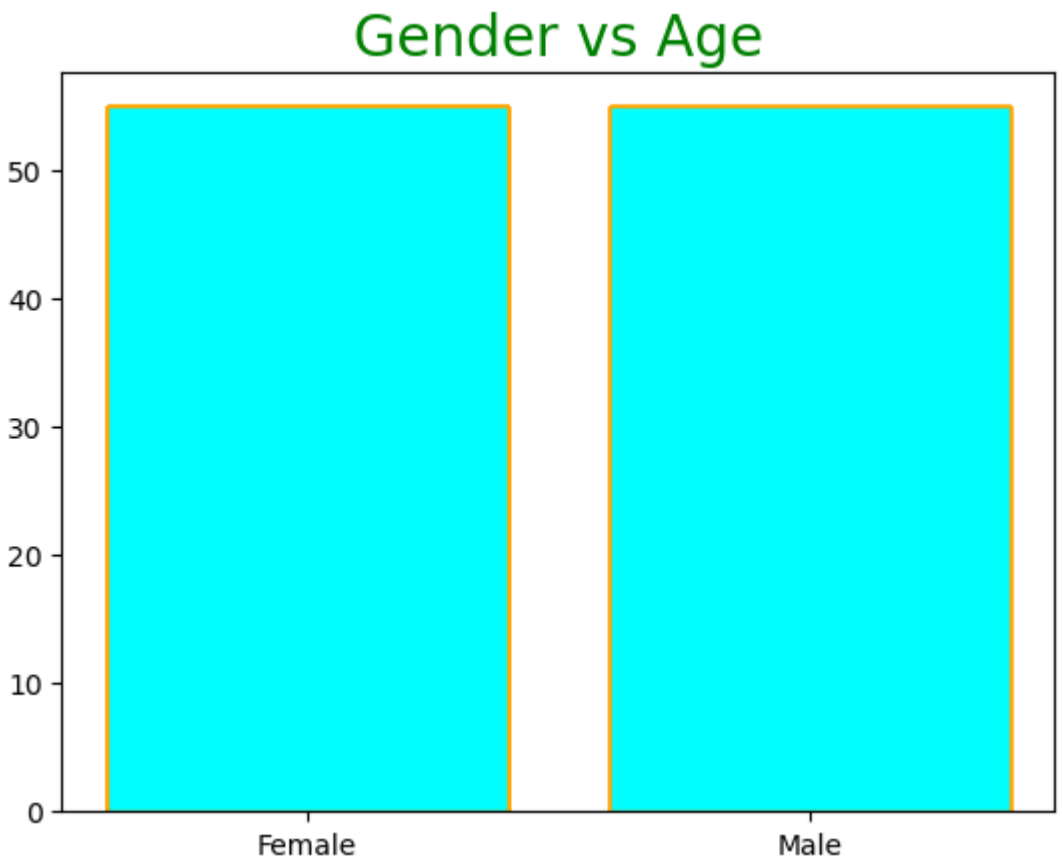
In [63]:
```
# Comparing two entities
plt.bar(churn_df['Geography'], churn_df['EstimatedSalary'].mean(), color="cyan",
plt.title('Geography vs Estimated Salary')
plt.show()
```



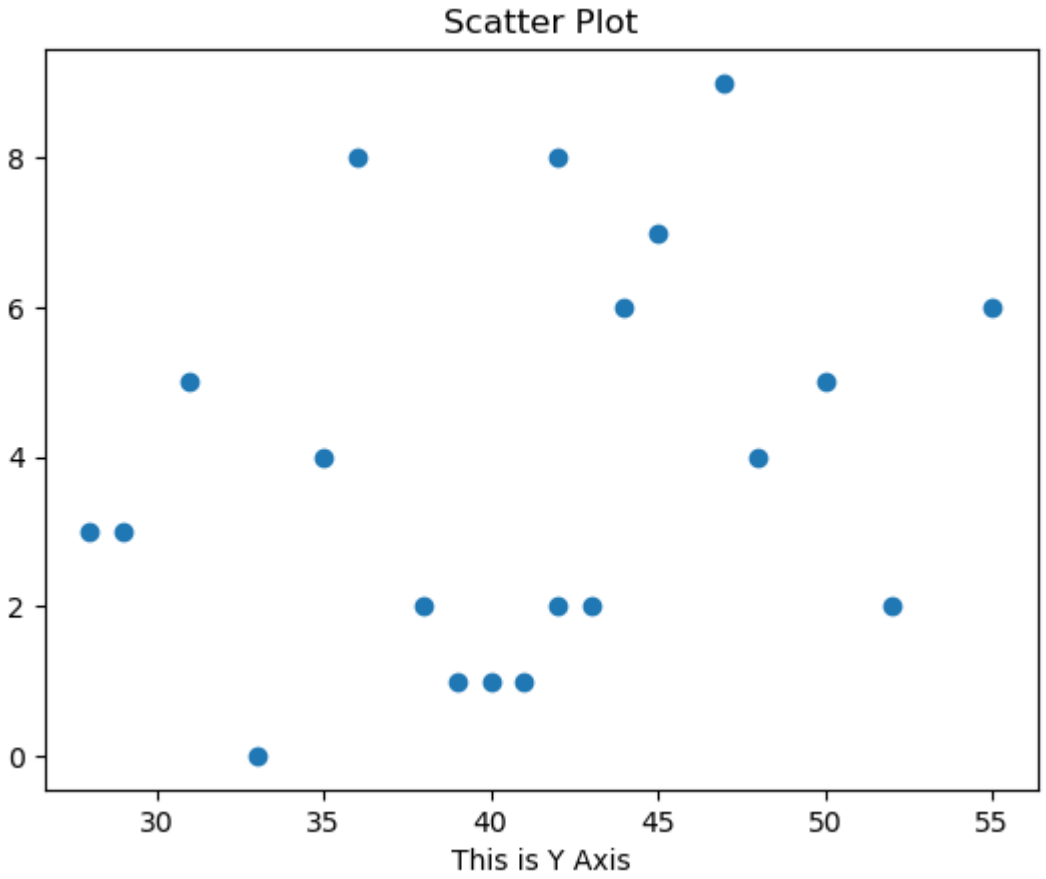
In [64]:
```
plt.bar(churn_df['Geography'], churn_df['EstimatedSalary'].max(), color="cyan",
plt.title('Geography vs Estimated Salary', fontdict = {"fontsize": 20, "color":"
plt.show()
```



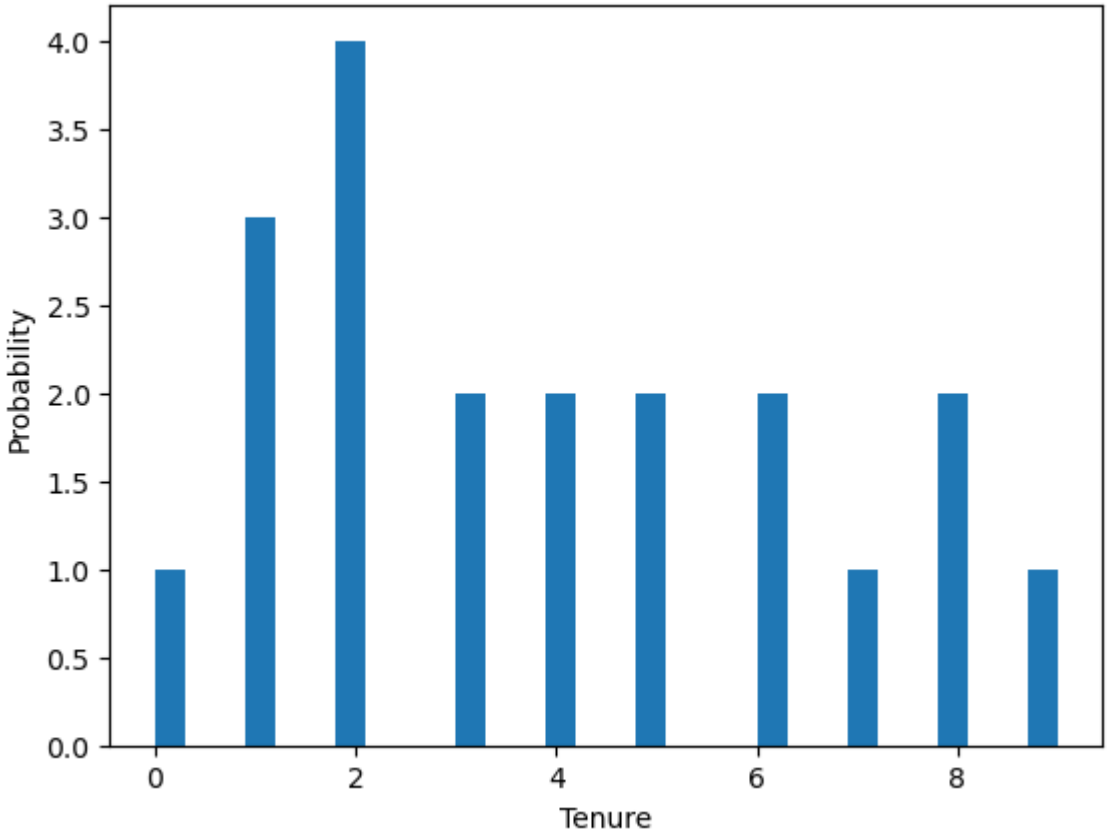
In [65]:
```
plt.bar(churn_df['Gender'], churn_df['Age'].max(), color="cyan", edgecolor="oran
plt.title('Gender vs Age', fontdict = {"fontsize": 20, "color":"Green"})
plt.show()
```



In [68]:
```
# When we analysis multiple numerical columns then we analyse using scatter plot
# It helps to understand what is the relationship between various attributes
plt.scatter(churn_df['Age'], churn_df['Tenure'])
plt.title('Scatter Plot')
plt.xlabel('This is X Axis')
plt.xlabel('This is Y Axis')
plt.show()
```

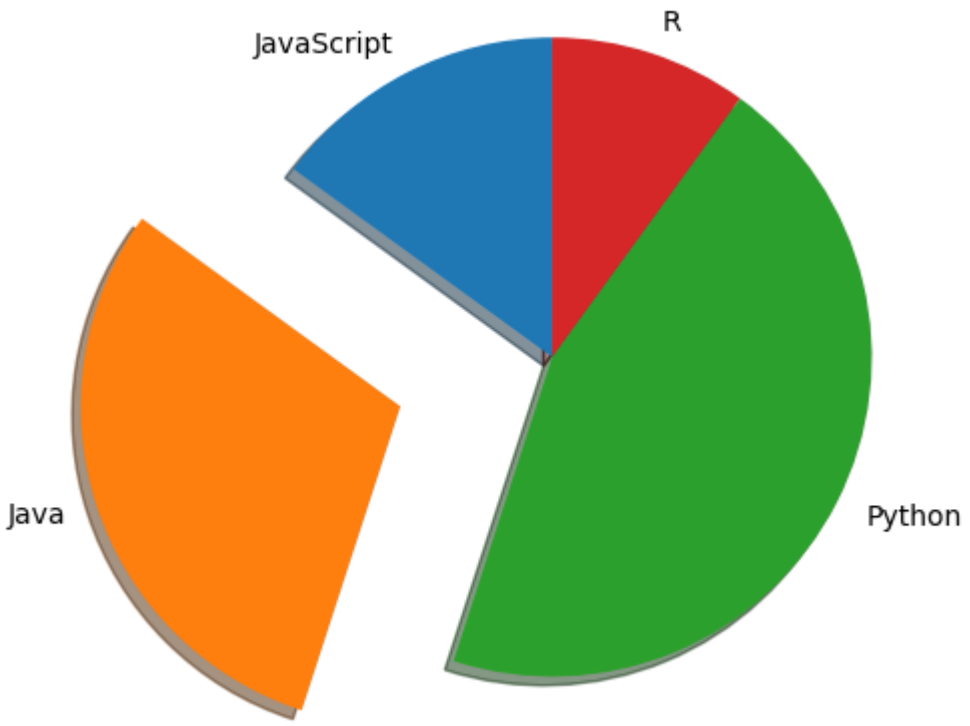


In [67]:
```
plt.hist(churn_df['Tenure'], bins=30)
plt.ylabel('Probability')
plt.xlabel('Tenure')
plt.show()
```

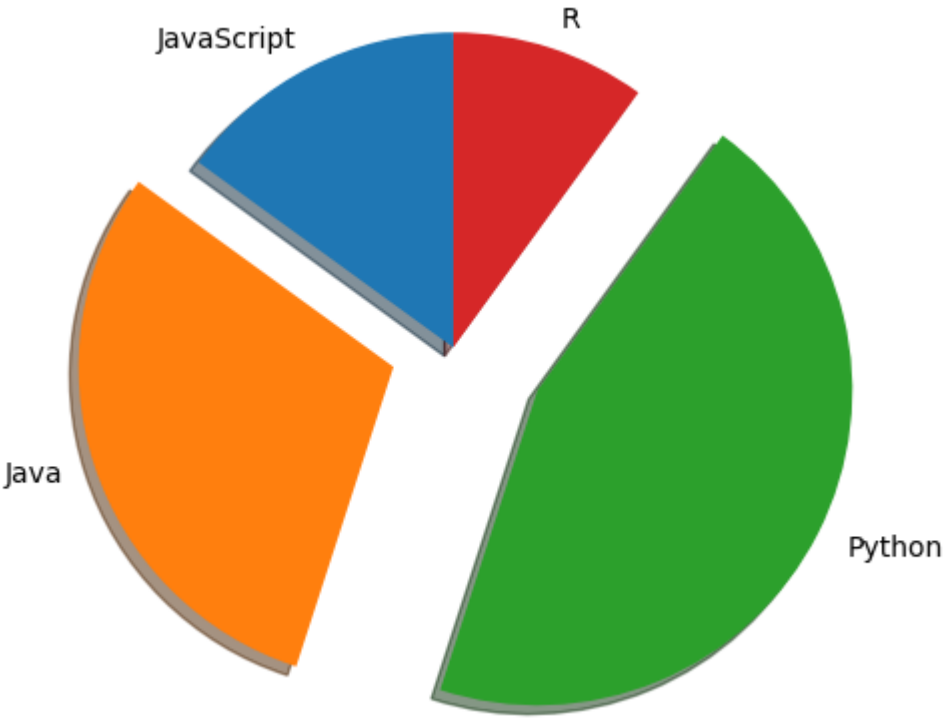


In [70]:
```
labels_ex = 'JavaScript', 'Java', 'Python', 'R'
```

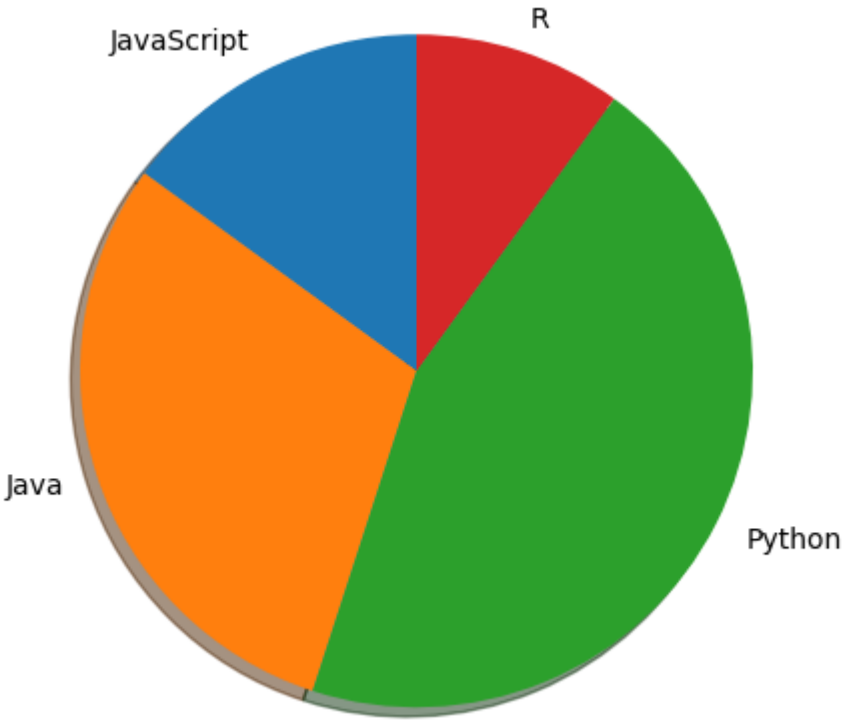
```
sizes = [15,30,45,10]
explode_labels= (0,0.5,0,0)
fig1, ax1 = plt.subplots()


ax1.pie(sizes, explode = explode_labels, labels=labels_ex, shadow=True, startang
ax1.axis('equal')
plt.show()
```



In [72]:
```
labels_ex = 'JavaScript', 'Java', 'Python', 'R'
sizes = [15,30,45,10]
explode_labels= (0,0.2,0.3,0)
fig1, ax1 = plt.subplots()


ax1.pie(sizes, explode = explode_labels, labels=labels_ex, shadow=True, startang
ax1.axis('equal')
plt.show()
```

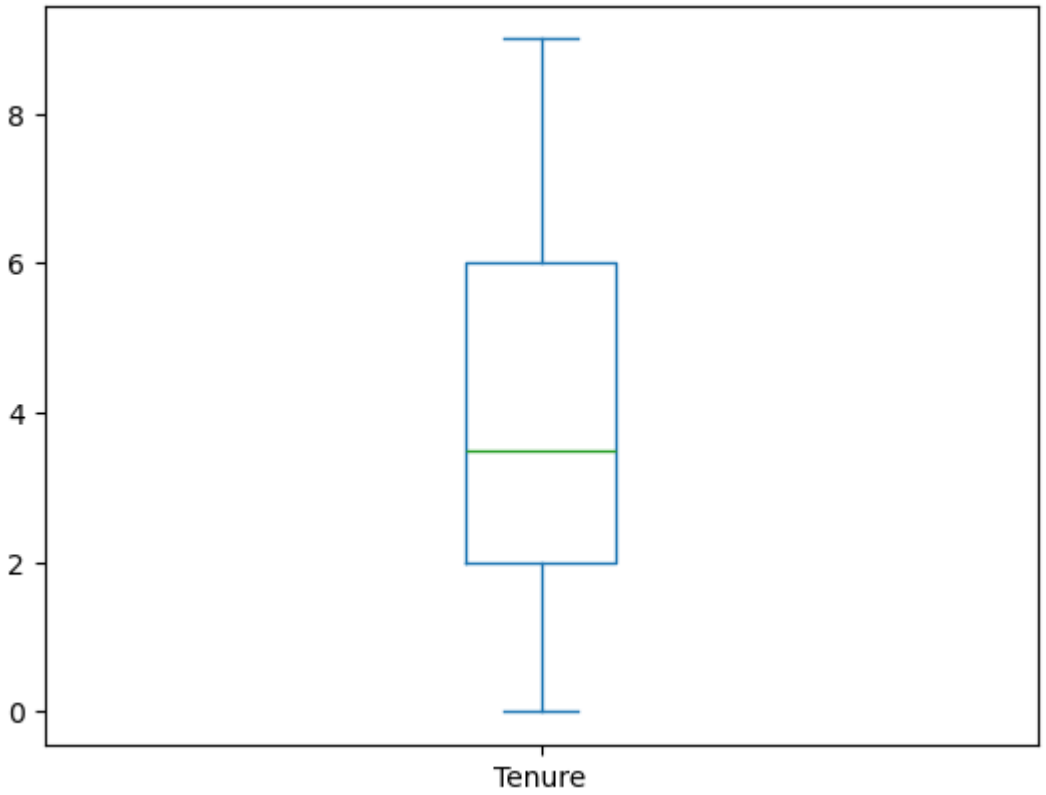


In [71]:
```
labels_ex = 'JavaScript', 'Java', 'Python', 'R'
sizes = [15,30,45,10]
explode_labels= (0,0,0,0)
fig1, ax1 = plt.subplots()


ax1.pie(sizes, explode = explode_labels, labels=labels_ex, shadow=True, startang
ax1.axis('equal')
plt.show()
```



In [73]:
```
churn_df['Tenure'].plot.box()
```

Out[73]: <Axes: >



In [74]:
```
churn_df['Tenure'].describe()
```

Out[74]:
```
count    20.000000
mean      3.950000
std       2.684752
min       0.000000
25%       2.000000
50%       3.500000
75%       6.000000
max       9.000000
Name: Tenure, dtype: float64
```